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On financial risk and the safe haven characteristics of Swiss franc exchange rates

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ABSTRACT

We analyse bilateral Swiss franc exchange rate returns in an asset pricing framework to evaluate the Swiss franc's safe haven characteristics. A "safe haven" currency is a currency that offers hedging value against global risk, both on average and in particular in crisis episodes. To explore these issues we estimate the relationship between exchange rate returns and risk factors in augmented UIP regressions, using recently developed econometric methods to account for the possibility that the regression coefficients may be changing over time. Our results highlight that in response to increases in global risk the Swiss franc appreciates against typical carry trade investment currencies such as the Australian dollar, but depreciates against the US dollar, the Yen and the British pound. Thus, the Swiss franc exhibits safehaven characteristics against many, but not all other currencies. We find statistically significant time variation in the relationship between Swiss franc returns and risk factors, with this link becoming stronger in times of stress.

1. Introduction

The recent crisis in the euro area has led to a massive appreciation of the Swiss franc, prompting the Swiss National Bank (SNB) to implement unconventional policy measures, including foreign exchange interventions and the introduction of an exchange rate floor against the euro.[1] In this context, the usual explanation put forward for the strong Swiss franc appreciation is the status of the Swiss franc as the typical safe haven currency.[2] Safe haven assets provide a hedge against risk on average. This characteristic is amplified in severe crises episodes during which safe haven assets particularly gain in value. High frequency analysis of Swiss franc exchange rate movements indeed leaves the impression of safe haven characteristics of the Swiss franc in several crises events (Ranaldo and Söderlind, 2010).

In this paper we argue that the Swiss franc exhibits safe haven asset characteristics against some currencies but not against other major currencies, such as the US dollar and the Yen. We draw this conclusion from studying Swiss franc exchange rate changes in an asset pricing framework, using recently developed econometric methods to assess time variation in the relation between exchange rates and risk factors. A steadily growing literature argues that (ex post) deviations from the uncovered interest rate parity (UIP) condition can be rationalized by the covariation of exchange rate changes with risk factors (e.g. Lustig

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[1] In the 3 years leading up to the introduction of the exchange rate floor against the euro in September 2011 the Swiss franc appreciated almost 40% against the euro.
[2] See Habib and Stracca (2012) for an empirical assessment of the factors that determine a safe haven.

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and Verdelhan, 2006, 2007; Ranaldo and Söderlind, 2010; Verdelhan, 2010, 2012; Farhi and Garbaix, 2011; Lustig et al., 2011; Menkhoff et al., 2012; Sarno et al., 2012; Bansal and Shaliastovich, 2013). Safe haven characteristics imply a close link to financial risk factors. It is hence natural to analyse Swiss franc exchange rate changes in this framework. We adopt the asset pricing framework of Verdelhan (2012), based on Backus et al. (2001) and Lustig et al. (2011), to analyse 11 bilateral Swiss franc exchange rate pairs during the time period from January 1990 to August 2011. This framework features one Swiss franc-specific and one global risk factor.

Consistent with Lustig et al. (2011) and Verdelhan (2012) we find that exposure to the Swissfranc -specific risk factor explains most of the time variation in Swiss franc returns. However, it is the sensitivity to global risk that reveals the safe haven characteristic of a currency. A safehaven currency gains in value (appreciates) relative to other currencies when global risk, risk that affects all currencies, materializes. Our results highlight that the Swiss franc is indeed a safe haven relative to many, but not all currencies: in response to increases in global risk the franc appreciates against typical carry trade investment currencies such as the Australian dollar, but depreciates against the US dollar, the Yen and the British pound. Exploiting insights of Müller and Petalas (2010) on the estimation of time-varying regression coefficients we find statistically significant time variation in the relationship between Swiss franc returns and risk factors, with this link becoming stronger in times of stress. For instance, on average a one percent increase in the VIX index — our baseline proxy for global risk — is associated with 0.04% Swiss franc appreciation against the Australian dollar, and a 0.03% depreciation against the US dollar. Around the period of the Lehman bankruptcy, the change in the VIX index was associated with a more than 0.2% appreciation against the Australian dollar, and a more than 0.2% depreciation against the US dollar.

The remainder of the paper is organized as follows. Section 2 provides information about the conceptual background of this study. Section 3 describes the data sources. Section 4 presents the econometric framework and the main empirical results. Section 5 concludes.

2. Conceptual background

This section motivates the use of an asset pricing framework to explain exchange rate changes. It provides some basic, theoretical background and introduces recent advances in the formulation of empirical currency risk models that form the backbone of our empirical analysis.

2.1. UIP regressions

UIP states that under the assumption of rational expectations and risk-neutrality the expected exchange rate change reflects the previous period's interest rate differential between the home and foreign country, i.e.

$$E_t\left(s^k_{t+1}\right) - s^k_t = i^k_t - i_t + \xi_{t+1} \tag{1}$$

where ξ_{t+1} is a risk premium, i^k_t is the country k interest rate, i_t its home country counterpart, s^k_{t+1} the log spot exchange rate of the home country relative to country k and E is the expectation operator. An increase in s corresponds to an appreciation of the home and depreciation of the foreign (country k) currency.

Interest rate differentials are approximately equal to forward discounts at least at the monthly frequency that we consider (e.g. Akram et al., 2008), such that

$$i^k_t - i_t \approx f^k_t - s^k_t \tag{2}$$

with f^k_t the log forward exchange rate. Under the assumption or rational expectations we have

$$E_t\left(s^k_{t+1}\right) = s^k_{t+1} + u^k_{t+1} \tag{3}$$

where the forecast error u^k_{t+1} is white noise. In particular, u^k_{t+1} is uncorrelated with any information that is available in period t. Substituting this expression for the expectation of the future spot rate in Eq. (1) gives

$$\Delta s^k_{t+1} = \left(f^k_t - s^k_t\right) + \xi_{t+1} + u^k_{t+1}. \tag{4}$$

The standard UIP regression for the bilateral exchange rate with country k then has the following form:

$$\Delta s^k_{t+1} = \alpha^k + \beta^k\left(f^k_t - s^k_t\right) + \varepsilon^k_{t+1}. \tag{5}$$

According to the UIP condition, the regression coefficient β should be equal to unity and the constant term, α, should be equal to zero. The error term ε^k_{t+1} reflects both forecast errors and the risk premium.

However, starting with Tryon (1979), Hansen and Hodrick (1980) and Fama (1984), support for the UIP condition remains rather scarce.[3] Ex post there are deviations from UIP. One potential explanation for the lack of empirical support for UIP is the presence of a risk premium required by market participants for a foreign currency investment (e.g. Lustig and Verdelhan, 2006, 2007; Ranaldo and Söderlind, 2010; Verdelhan, 2010, 2012; Farhi and Garbaix, 2011; Lustig et al., 2011; Menkhoff et al., 2012; Sarno et al., 2012; Bansal and Shaliastovich, 2013). According to this literature, covariation of exchange rate returns with contemporaneous currency risk factors is responsible for the ex post deviation from the UIP condition. We briefly highlight the theoretical background of this strand of the literature in the subsequent section.

But a large literature uses survey-based expectations of exchange rates to show that UIP holds reasonably well ex ante, i.e. when we do not impose the assumption of rational expectations. Recent papers on this subject include Chinn (2014) and Bacchetta et al. (2009). Early contributions were made by Dominguez (1986) and Frankel and Froot (1987). We find similar evidence for the Swiss franc exchange rates under study and report these results as robustness check in Section 4.3. This finding suggests that the asset pricing view on currency returns is not necessarily the best or only explanation of ex post deviations from UIP. From a different angle Hassan and Mano (2014) make a similar point by stressing that ex post deviations from UIP and returns on currency investment strategies are separate phenomena. We corroborate these points.

However, asset pricing models for currency returns have a strong appeal for the assessment of currency investment strategies and the safe haven characteristics of exchange rates because a safe haven currency is a currency that offers investors hedging value against global risk. This assessment is the contribution of this paper.

2.2. Pricing models of exchange rate returns: theoretical background

Asset pricing models of exchange rate returns interpret the UIP condition as a (zero net) investment strategy. At time t, this strategy first requires borrowing at the home country interest rate i and converting the home currency at the spot exchange rate into $1/S$ foreign currency units (FCUs). The FCUs are invested at the foreign interest rate i^k. At time $t+1$ the investor converts the FCUs into the home currency at the spot rate S_{t+1} (Burnside et al., 2011). This zero net investment strategy leads to the following payoff, x_{t+1}:

$$x_{t+1} = \frac{1}{S_t}\left(1+i_t^k\right)S_{t+1} - (1+i_t). \tag{6}$$

Having interpreted the UIP condition in this manner allows directly applying the basic asset pricing equation for excess returns (Cochrane, 2005), such that

$$E_t\left[\left(\frac{1}{S_t}\left(1+i_t^k\right)S_{t+1} - (1+i_t)\right)m_{t+1}\right] = 0 \tag{7}$$

should hold in which m_{t+1} denotes the stochastic discount factor.

Dividing by $E_t(m_{t+1})$ and rearranging yields the "risk-adjusted" version of the UIP condition, i.e.

$$(1+i_t) = \left(1+i_t^k\right)\left[E_t\left(\frac{S_{t+1}}{S_t}\right) + \frac{\operatorname{cov}\left[(S_{t+1}/S_t), m_{t+1}\right]}{E_t(m_{t+1})}\right]. \tag{8}$$

If we combine Eq. (8) with the covered interest rate parity (CIP) condition

$$(1+i_t) = \frac{1}{S_t}\left(1+i_t^k\right)F_t \tag{9}$$

where F_t abbreviates the forward exchange rate at time t, then, after rearranging, we obtain that version of the "risk-adjusted" UIP condition that underlies most of the empirical studies of the link between risk factors and exchange rate returns:

$$E_t\left(\frac{S_{t+1}-S_t}{S_t}\right) = \frac{F_t - S_t}{S_t} - \frac{\operatorname{cov}\left[m_{t+1}, ((S_{t+1}-S_t)/S_t)\right]}{E_t(m_{t+1})}. \tag{10}$$

According to Eq. (8), covariation with the stochastic discount factor influences expected exchange rate returns on top of the previous period's forward discount/interest rate differential. This rationale forms the backbone of recently proposed asset pricing models for exchange rate returns that we apply in the empirical part of this paper and that we describe in the next subsection.

[3] Bansal and Dahlquist (2000) find that UIP holds for high inflation countries. Meredith and Chinn (2005) use long-term government bond yields as a proxy for risk-free rates to assess UIP at long horizons. They find that UIP holds quite well for horizons of 5 years or more. Lothian and Wu (2005) provide evidence that UIP holds over the long-term if we exclude the 1980s from the sample. Huisman et al. (1998) use a panel setup to show that UIP is violated but with significant, non-negative regression coefficients.

2.3. Pricing models of exchange rate returns: recent empirical advances

According to this asset pricing view on exchange rate determination, UIP regressions, such as Eq. (5), should be augmented by incorporating currency risk factors (Verdelhan, 2012). Such regressions could then take the following form

$$\Delta s_{t+1}^{k} = \alpha^{k} + \beta_0^{k}\left(f_t^{k} - s_t^{k}\right) + \beta_1^{k}\psi_{t+1}^{1} + \beta_2^{k}\psi_{t+1}^{2} + \ldots + \beta_n^{k}\psi_{t+1}^{n} + \varepsilon_{t+1}^{k} \tag{11}$$

where n denotes the numbers of risk factors in a currency risk premium model, k represents currency k and ψ denotes a currency risk factor.

Note that such regressions additionally assume that the investor's discount factor is a linear function of the risk factors ψ. Moreover, these risk factor augmented UIP regressions assess exchange rate returns (or, in general, asset returns) in "risky" or "less risky" times as measured by the factors. Therefore, the exchange rate returns are regressed on contemporaneously measured factors (Verdelhan, 2012).

We know at least since Meese and Rogoff (1983) that exchange rate changes are notoriously hard to predict. The asset pricing framework under study does not alter this conclusion. We remain agnostic whether this indeed constitutes a major puzzle for international finance (Bacchetta and van Wincoop, 2006) or whether it should not be too surprising (Engel and West, 2005).

How do we approximate currency risk factors empirically? Since standard risk factors (e.g. the factors proposed by Fama and French, 1993) fail to describe exchange rate returns (Burnside et al., 2011), we follow recent studies that extract information about currency risk factors directly from exchange rate data. Lustig et al. (2011) show that the two first principal components in portfolios of foreign currency risk premia, sorted according to currencies' forward discounts (interest rate differentials), correspond to a country-specific and a global risk factor of currency excess returns (ex post deviations from UIP) from the U.S. investor's perspective. Differences in the exposure to the global factor determine the average risk premium on the currency portfolios. The exposures of these currency portfolios to the country-specific factor are almost equal. Lustig et al. (2011) further show that their empirical model reflects an extension of the two-country framework of Backus et al. (2001) to a multi-country, global context. In this model, the idiosyncratic factors play no role in the determination of currency risk premia. By forming portfolios of foreign currencies, Lustig et al. (2011) are able to ensure empirically that this is the case, i.e. average out idiosyncrasies of currency excess returns.

In addition, Lustig et al. (2011) in the context of currency excess returns and Verdelhan (2012) in the context of exchange rate changes show that these risk factors, extracted from portfolios of currency excess returns, are also informative about time series as well as cross-sectional variation in bilateral excess returns and bilateral exchange rate changes. Therefore we adopt this model as our baseline specification to study Swiss franc exchange rate returns.

3. Data

Our baseline sample period spans the time period from January 1990 to August 2011. The beginning of the sample period is limited by the availability of data for our proxies of (global) currency risk, while the end of the sample reflects the introduction of the minimum rate against the euro by the SNB in September 2011. We consider Swiss franc exchange rates relative to the Australian dollar, the Canadian dollar, the euro, the Japanese yen, the New Zealand dollar, the Norwegian krone, the Singaporean dollar, the South African rand, the Swedish krona, the British pound and the US dollar. Before the introduction of the euro in January 1999, we use the European Currency Unit (ECU) as proxy for the euro exchange rate. We obtain these exchange rates from cross-rates of US dollar exchange rates. The source of the respective end-of-month spot exchange rate data and the one-month forward rates are Barclays and WM/Reuters, available via Datastream. The sample of exchange rates is limited by the availability of exchange rate data over the entire sample period.

We focus on data at the monthly frequency because the augmented UIP regressions feature forward discounts as approximation of interest rate differentials. Hence, we assume that covered interest rate parity holds. This relation only holds at weekly or lower frequencies (e.g. Akram et al., 2008). Despite this limitation the use of forward discounts instead of interest rate differentials has some advantages. First, forward discounts are actually traded while money market rates, as e.g. Libor rates, do not necessarily reflect actual transactions. Second, UIP requires that interest rates reflect the same "riskiness". Again, the use of money market rates or government bond yields could lead to violations of this requirement.

Moreover, we employ changes and levels in the CBOE option-implied volatility index of the S&P 500, VIX, based on end of month close prices as a proxy of global currency risk. The VIX series is obtained directly from the CBOE website. As an additional check for euro area specific risks we use the yield spread (in percentage points p.a.) between Italian and German government bonds with constant 10-year maturity from Datastream. The sample period for this data is April 1991 to August 2011. Furthermore, we employ the TED spread, i.e. the spread between a 3-month US Treasury bill and the 3-month Eurodollar deposit rate in percentage points p.a., as measure of funding liquidity risk and hence as additional control variable in our robustness checks (Brunnermeier et al., 2009).

4. Econometric framework and empirical results

Taking currency risk factors into account enhances our understanding of exchange rate dynamics. Section 4.1 provides evidence supporting that point. Section 4.2 presents evidence of considerable time variation in the exchange rate–risk factor relation. We provide a summary of various robustness checks in Section 4.3.

4.1. (Augmented) UIP regressions

This section highlights the usefulness of the asset pricing view of exchange rates for studying Swiss franc returns. We start by showing the results of standard UIP regressions for the Swiss franc which provide the motivation to use empirical models augmented by currency risk factors. The second subsection presents the results for these models.

4.1.1. UIP regressions

This section assesses if the uncovered interest rate parity condition holds ex post in the context of the Swiss franc exchange rates under study. Table 1 provides estimates of α^k and β_0^k from monthly UIP regressions

$$\Delta s_{t+1}^k = \alpha^k + \beta_0^k \left(f_t^k - s_t^k\right) + \varepsilon_{t+1}^k \tag{12}$$

where k denotes one of bilateral Swiss franc exchange rates and s and f represent log spot exchange rate changes and 1-month log forward rates respectively. Newey–West (Newey and West, 1987) corrected standard errors appear below the point estimates in parentheses. The R^2 statistic is adjusted for the number of regressors. The Durbin–Watson statistic (DW) indicates if there is autocorrelation in the regression residuals. The sample period runs from January 1990 to August 2011.

The results summarized in Table 1 show that UIP is violated ex post for about half of the currency pairs under study. In these cases, which include all of the major currencies such as the Yen or the US dollar, we reject the null of $\beta_0^k = 1$. By contrast, we cannot reject that UIP holds ex post in the other half of the sample. Consider for example, the Australian dollar vis-à-vis the Swiss franc. The point estimate of β_0^k is 0.13 with a standard error of 1.29. The corresponding t-statistic of the null that this regression coefficient is different from unity is hence $(0.13 - 1) / 1.29 = -0.67$.

However, the asterisks in Table 1 indicate that there are only two coefficient estimates of β_0^k that are statistically different from zero. In sum, the standard UIP regressions suggest that lagged forward discounts/interest rate differentials explain little of the time variation in Swiss franc exchange rate returns. The respective measures of fit range from 0% to 4%.

As we highlighted in Section 2.2, covariation with contemporaneously measured risk factors could be an additional determinant of exchange rate returns. Hence, we augment the standard UIP regression with currency risk factors and evaluate the potential covariation of the Swiss franc exchange rate returns with these risk factors in the subsequent section.

4.1.2. Augmented UIP regressions

In this subsection, we assess if the consideration of potential currency risk factors helps to better understand exchange rate dynamics. In our baseline specification, we closely follow Verdelhan (2012) and Lustig et al. (2011) and adopt their asset pricing

Table 1
UIP regressions.

	α^k	β_0^k	R^2	DW
AUD	−0.00 (0.01)	0.13 (1.29)	0.00	1.94
CAD	−0.00 (0.00)	− 0.75 (−1.37)	0.00	2.11
EUR	−0.00 (0.00)	−1.54 (1.15)	0.01	2.29
JPY	0.00 (0.00)	−1.71* (0.72)	0.01	1.98
NZD	−0.00 (0.00)	−0.12 (0.89)	0.00	1.90
NOK	−0.00 (0.00)	−0.96 (1.05)	0.01	2.21
SGD	−0.00 (0.00)	0.56 (0.86)	0.00	1.95
ZAR	−0.01* (0.00)	−0.42* (0.10)	0.04	1.99
SEK	−0.00 (0.00)	1.07 (0.66)	0.01	2.10
GBP	−0.01* (0.00)	−1.30 (0.83)	0.00	1.99
USD	−0.00 (0.00)	−1.54 (1.09)	0.01	1.91

Notes: This table presents estimates from regressions of Swiss franc exchange rate returns (Swiss franc relative to Australian dollar, Canadian Dollar, euro, Japanese yen, New Zealand dollar, Norwegian krone, Singaporean dollar, South African rand, Swedish krona, British pound and US dollar) on a constant and the respective forward discount, i.e.

$$\Delta s_{t+1}^k = \alpha^k + \beta_0^k \left(f_t^k - s_t^k\right) + \varepsilon_{t+1}^k$$

where k denotes the currency pairs and s and f represent log spot exchange rates and 1-month log forward rates, respectively. Newey–West (Newey and West, 1987) corrected standard errors appear below the point estimates in parentheses. The R^2 statistic is adjusted for the number of regressors. An asterisk indicates that estimates are significantly different from zero at 95% confidence level. We report the Durbin–Watson statistic for the autocorrelation of the regression residuals under the column "DW". The sample period runs from January 1990 to August 2011.

framework to the Swiss franc exchange rate context. We consider different specifications as robustness checks and summarize these results in a separate, subsequent subsection.

Our baseline currency pricing model consists of two risk factors (Lustig et al., 2011). The first factor is the average Swiss franc exchange rate change, abbreviated with AFX. It is a currency-specific factor as shown by Lustig et al. (2011). This factor accounts for most of the time variation in currency (excess) returns. We exclude the exchange rate whose dynamics we evaluate from the calculation of the currency-specific factor as recommended by Verdelhan (2012). For example, when we examine the Swiss franc–euro exchange rate changes in the augmented UIP regressions, we use the arithmetic average of the other 10 Swiss franc exchange rate changes as first risk factor. Verdelhan (2012) shows that this average of exchange rate changes indeed corresponds to a risk factor on currency markets. He shows that, from the perspective of a US investor, returns on currencies sorted in portfolios according to their exposure to this factor also reveal strong cross-sectional differences.

Of course, as is common in the currency risk factor literature, we have to live with the fact that the number of exchange rates is limited. Unlike, for example, the market return in the standard CAPM, AFX thus has to be constructed from a relatively small cross-section of currencies.

As a second risk factor, we employ returns on the VIX index. This second risk factor is a measure of global risk on currency markets. We exploit that the empirical proxy of the global risk factor by Lustig et al. (2011), differences in the returns on high and low forward discount sorted currency baskets, is positively correlated with innovations in global equity market volatility. VIX returns are our proxy of innovations in global equity market volatility and hence our approximation of the global factor in the original Lustig et al. (2011) model. This VIX index is also highly positively correlated with volatility indices of other equity markets. The use of this equity market volatility series as proxy of global risk helps to avoid potential double counting in the augmented UIP regressions. The global factor in Lustig et al. (2011) is constructed from currency portfolios sorted on forward discounts/interest rate differentials. Hence, including this factor directly in a regression together with the bilateral forward discount rate of the exchange rate under study could create econometric issues since that particular currency pair or a cross rate is included in the Lustig et al. (2011) global risk factor. We therefore prefer to use a global risk factor proxy that is neither constructed from interest rate differentials/forward discounts nor from exchange rate data.

As Lustig et al. (2011) have shown, cross-sectional differences in foreign currency returns are mainly due to differences in the sensitivities of these returns to the global risk factor. Hence, exposures to global risk should reflect safe haven characteristics of a currency: a safe haven currency should have negative exposure to the global risk factor, i.e. it should gain in value (appreciate) when global risk materializes and thus provide a hedge for all investors.

The augmented UIP regressions then take the following form

$$\Delta s^k_{t+1} = \alpha^k + \beta^k_0\left(f^k_t - s^k_t\right) + \beta^k_1 \mathrm{AFX}_{t+1} + \beta^k_2 \Delta \log(\mathrm{VIX})_{t+1} + \varepsilon^k_{t+1} \tag{13}$$

Table 2
Augmented UIP regressions.

	α^k (Constant)	β^k_0 ($f^k_t - s^k_t$)	β^k_1 (AFX_{t+1})	β^k_2 ($\Delta\log(\mathrm{VIX})_{t+1}$)	R^2	DW
AUD	0.00 (0.00)	0.26 (0.63)	1.29* (0.09)	−0.04* (0.01)	0.65	2.01
CAD	0.00 (0.00)	−0.55 (0.55)	1.37* (0.05)	−0.00 (0.00)	0.71	2.24
EUR	−0.00 (0.00)	−0.95 (0.93)	0.35* (0.05)	−0.01 (0.00)	0.33	2.19
JPY	0.00 (0.00)	−1.44* (0.70)	0.64* (0.10)	0.03* (0.01)	0.18	1.91
NZD	0.00 (0.00)	0.09 (0.72)	1.04* (0.10)	−0.02* (0.01)	0.50	1.98
NOK	−0.00 (0.00)	−0.31 (0.82)	0.59* (0.10)	−0.01 (0.01)	0.36	2.14
SGD	0.00 (0.00)	0.71* (0.31)	1.09* (0.06)	0.02* (0.01)	0.73	1.84
ZAR	−0.01* (0.00)	−0.39* (0.10)	1.08* (0.10)	−0.05* (0.01)	0.43	2.06
SEK	−0.00 (0.00)	0.22 (0.70)	0.54* (0.09)	−0.02* (0.01)	0.30	1.90
GBP	−0.00 (0.00)	−0.78 (0.53)	0.81* (0.11)	0.02* (0.01)	0.42	1.97
USD	−0.00 (0.00)	−0.23 (0.56)	1.20* (0.06)	0.03* (0.01)	0.60	1.70

Notes: This table presents estimates from regressions of Swiss franc exchange rate changes (Swiss franc relative to Australian dollar, Canadian Dollar, euro, Japanese yen, New Zealand dollar, Norwegian krone, Singaporean dollar, South African rand, Swedish krona, British pound and US dollar) on a constant, the respective forward discount and two contemporaneously measured risk factors, i.e.

$$\Delta s^k_{t+1} = \alpha^k + \beta^k_0\left(f^k_t - s^k_t\right) + \beta^k_1 \mathrm{AFX}_{t+1} + \beta^k_2 \Delta \log(\mathrm{VIX})_{t+1} + \varepsilon^k_{t+1}$$

with AFX abbreviating average Swiss franc exchange rate returns excluding currency k and Δ log(VIX) representing log changes in the CBOE option-implied volatility index, VIX. k denotes the currency pairs and s and f represent log spot exchange rate and 1-month log forward rates respectively. Newey–West (Newey and West, 1987) corrected standard errors appear below the point estimates in parentheses. The R^2 statistic is adjusted for the number of regressors. An asterisk indicates that estimates are significantly different from zero at 95% confidence level. We report the Durbin–Watson statistic for the autocorrelation of the regression residuals under the column "DW". The sample period runs from January 1990 to August 2011.

with AFX abbreviating average Swiss franc exchange rate changes excluding currency k and Δ log(VIX) representing log changes in the CBOE option-implied volatility index, VIX. The regression results are summarized in Table 2. Newey–West (Newey and West, 1987) corrected standard errors appear below the point estimates in parentheses. The sample period runs from January 1990 to August 2011.

Several observations are noteworthy. First, the coefficients on the forward discount remain in most cases not statistically different from zero. Second, we observe pronounced variation in the sensitivity to the Swiss franc-specific currency risk factor, AFX. The estimates range from 1.4 for the Canadian dollar to 0.4 for the euro. One potential reason why Swiss franc–euro returns are less strongly related to Swiss franc movements against other currencies may be the fact that the Swiss economy is closely linked to the euro area, dominating the effects from the "rest-of-the-world". Recall that the respective AFX series do not include the bilateral exchange rates that we evaluate in the respective regression. Third, the covariation of Swiss franc exchange rates with VIX returns shows that the Swiss franc exhibits safe haven characteristics against many, but not against all other currencies. The coefficient β_2^k reflects both the Swiss franc's status as a safe haven and the franc's role as a typical funding currency in currency carry trades. Negative coefficient estimates indicate that the Swiss franc appreciates against the respective currency when the VIX, i.e. global risk, increases. This is the case for the Swiss franc exchange rates against the typical investment currencies in carry trades such as the Australian dollar, the New Zealand dollar or the South African rand. By contrast, significant positive coefficients signal that the Swiss franc depreciates against the respective currency when global risk increases. This is the case for the Swiss franc exchange rates against the Japanese Yen, the British pound and the US dollar. On average, these currencies provide a better hedge against global risk than the Swiss franc.

A potential, general critique of our approach is that there could be structural breaks during the sample. In particular, one might argue that the foreign exchange interventions undertaken by the SNB since 2009 have changed the underlying "true" asset pricing relationship that determines Swiss franc exchange rate changes, and that ideally one should somehow control for SNB policy measures, for example by including a measure of the size of interventions as an additional control variable in the regression. However, the relevance of uncovered interest parity and the underlying risk-return relationship Eq. (13) for investors is unaltered by the presence of central bank interventions. Central bank interventions could, however, have changed the magnitude of the regression coefficients and thus have changed the exchange rate change–risk factor relation over time. We analyse potential time variation in the next section.[4]

4.2. Time variation in the exchange rate–risk factor relation

The previous section analysed the Swiss franc's safehaven characteristics by estimating the relationship between bilateral Swiss franc exchange rate returns and risk factors. However, there is evidence that this relation varies over time. For example, Christiansen et al. (2011) highlight that the relation between currency returns and more traditional risk factors such as the stock market return can be described in a regime-switching framework. The regime dependency is related to funding liquidity risk and volatility on foreign exchange markets.

It is intuitive that time variation in the exchange rate–risk factor relation is particularly important for safe haven assets. First, safe haven assets typically provide hedging value against risk events on average, but in particular also in times of financial stress (e.g. Ranaldo and Söderlind, 2010). Thus the comovement between returns on safehaven assets and risk factors can be expected to increase in financial crisis episodes (when risk factors are elevated). Second, as discussed above, the SNB's FX interventions in 2008–2011 could potentially have affected the link between Swiss franc movements and risk factors. To explore these issues we employ recently developed econometric methods to estimate the time variation in the coefficients of the baseline augmented-UIP regression (13), using log returns in the VIX index as the baseline measure of global risk. Our analysis is complementary to that of Ranaldo and Söderlind (2010): while they allow for nonlinearities in the relationship between safe haven assets and risk factors we instead estimate directly how this relationship has changed over time. [5]

We begin by reporting statistical tests for joint stability of the parameters in regression Eq. (13). To do this we employ the quasi-local-level test proposed by Elliott and Müller (2006).[6] The advantage of this test is that it is equivalent in large samples to the optimal tests for a wide range of possible statistical processes for time variation. Therefore we do not have to make specific assumptions about the particular process for time variation considered in the alternative hypothesis. The null of joint stability of the coefficients in Eq. (13) is rejected if the test statistic is smaller (more negative) than the critical values. We find that except for the CAD, SAR and GBP regressions the null of parameter stability can be rejected for all other currency pairs at the 5% level.

We then proceed to estimate the parameter paths for the coefficients, focusing on those bilateral rates where the null of parameter stability was rejected. To do this we use the method proposed by Müller and Petalas (2010) which provides an approximation to the

[4] In addition, we checked the pairwise correlations between the regressors and the residuals of the regressions. It turns out that the correlations between the regressors are relatively low (no pairwise correlation coefficient is higher than 0.3 in absolute value) and that the regression residuals are basically uncorrelated with the regressors.

[5] Hoffmann and Suter (2010) study five Swiss franc exchange rates in a conditional asset pricing framework over the period from 1990 to 2009. They regress these exchange rate returns on a global risk factor and an interaction term between the global risk factor and the respective currency pair's interest rate differential to study the Swiss franc's role as a safe haven. The interaction term captures the time variation in potential safe haven characteristics. Hoffmann and Suter (2010) argue that interest rate differentials are good signals of the safe haven status of a currency. While this is a natural starting point, Habib and Stracca (2012) point out that there are a couple of other macroeconomic variables that potentially signal the safe haven status of currencies. In addition, the importance of these macroeconomic variables as signals of safe haven status could also vary over time. The method that we use instead allows estimating directly how this relationship varies over time without relying on instrumental variables.

[6] This test has so far only been used in few applied papers, including Goldberg and Klein (2011).

sample information about the parameter path that is accurate and efficient in large samples, independent from the precise nature of the underlying process of time variation. Since we know from the currency risk factor literature that differences in the sensitivities to the proxy of global risk determine differences in exchange rate changes, we focus on the presentation of the time-varying path of the global risk measure coefficients. Fig. 1 reports the results for selected currency pairs.[7]

The largest movements in coefficients occur during financial crisis episodes, with coefficients peaking around the period of the Lehman bankruptcy and (for USD and JPY) during the period of the Asian–Russian financial crises in 1997–1998. Responses to movements in the risk factor are much larger in times of financial stress. For example, Table 2 reported that a 1% monthly increase in the VIX index is associated with a 0.04% appreciation of the Swiss franc versus the Australian dollar on average over the sample. In contrast, allowing for time variation shows that at the height of the global financial crisis in late 2008 a 1% increase in the VIX led to a 0.2% CHF appreciation against AUD, more than 5 times larger than the average effect. Confirming the findings from Table 2 for average comovements between Swiss franc returns and risk factors, during episodes of financial stress the negative relationship between Swiss franc returns against AUD, NZD and EUR and risk factors — with the Swiss franc appreciating in response to increases in the VIX index — becomes stronger: the Swiss franc's safehaven role is particularly pronounced during crisis episodes. Conversely, the Swiss franc depreciates more sharply against USD and JPY in response to increases in the risk factor when risk is elevated. Thus the finding from the baseline regressions that the Swiss franc can be considered a safe haven, in terms of its average hedging value, relative to EUR, AUD and NZD, but not relative to USD and JPY carries over to the definition of "safe haven" as an asset that gains in value in high-risk episodes.

This latter finding, especially for the USD, is economically intuitive. In the course of the dry-up of money markets and the financial turmoil around the Lehman bankruptcy, banks and other financial market participants scrambled for USD liquidity.[8] In addition, the USD is typically considered to be the global reserve currency with a deep and liquid market for USD denominated assets. Hence it is natural that it exhibits even stronger safe haven characteristics than the Swiss franc.

In addition, the time-variation in the sensitivity of the Swiss franc–Yen exchange rate return to VIX returns highlights that the safe-haven characteristic can vary over time. The fundamentals of the respective currency areas seem to play a vital role in this respect. While the Yen appreciated against the Swiss franc during the global financial crisis 2007–2009, it strongly depreciated around the time of the Asian crisis 1997/1998 when the VIX rose. In this time period, the Japanese economy fell into a recession and the Japanese banking sector was in a stressed situation (Corsetti et al., 1999). Against this background, it is natural that the Yen depreciated against the Swiss franc in this particular financial crisis period.

4.3. Robustness checks

Since the euro–Swiss franc exchange rate received particular attention by the SNB in the course of the euro area sovereign debt crisis, we present detailed robustness checks for this exchange rate in the next subsection. Thereafter, we use survey data on Swiss franc exchange rates to assess directly the importance of the assumption of rational expectations in the UIP regressions. Finally, we summarize various other robustness checks for which results are available upon request.

4.3.1. A closer look at the EURCHF in augmented UIP regressions

Given that the SNB introduced a minimum euro–Swiss franc exchange rate (EURCHF) in September 2011 to prevent a further massive appreciation of the Swiss franc, the regression results presented in the previous subsection seem to be surprising. The coefficient of the EURCHF return regressed on VIX returns is negative but statistically not different from zero.

One potential explanation for this finding is our use of ECU data to extend the sample period back to January 1990 in order to have comparable sample lengths for all currencies under study.

However, one can argue that the euro is fundamentally different from the ECU[9] such that the concatenation of ECU and euro data is not appropriate. Therefore, we run the regression Eq. (13) for euro only data. The sample period hence starts in January 1999. Table 3 reports the results.

To facilitate the comparison between the results from the previous subsection with the new regression results, we present again the regression results for the combination of ECU/euro data in panel A of Table 3. For expositional purposes we multiplied the regression coefficient of the exchange rate return (and the standard error) on the VIX return with 100. Panel B of Table 3 shows the results from the augmented UIP regression for the "true" euro sample period. In this regression, the coefficient of the EURCHF return on the VIX return almost doubles. It is negative — the Swiss franc appreciated relative to the euro when the VIX increased — and statistically different from zero. Hence, we find evidence that the Swiss franc exhibited safe haven characteristics relative to the euro.

This safe haven value of the Swiss franc relative to the euro even holds when we exclude the 2010/2011 sovereign debt crisis in the euro area from the sample period (as shown in panel C of Table 3). If we go further and also exclude the global financial crisis period, there is still a significant negative VIX return coefficient as the estimates presented in panel D of Table 3 show.

In sum, these regressions suggest that the Swiss franc has provided hedging value against global risks for an investor in EURCHF since the inception of the euro. It appreciated against the euro when global risks, as measured by VIX returns, rose.

[7] The time paths for the post-1999 period look almost unchanged if the estimation is performed for the shorter Euro sample (1999–2011) only.

[8] McCauley and McGuire (2009) list four factors that can potentially explain the USD safe haven characteristics during the crisis: flight to quality, unwinding of carry trades, a global dollar shortage, and overhedging.

[9] For example, in contrast to the ECU, the euro currency is controlled by a single central bank.

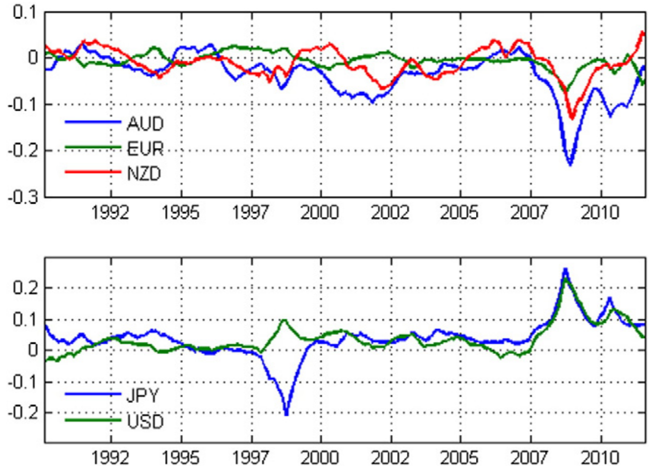

Fig. 1. Estimates of the parameter paths of the Δ log(VIX) coefficients from financial risk factor augmented UIP regression Eq. (13) for Swiss franc returns against the Australian dollar, the euro and the New Zealand dollar in the upper panel and against the Japanese Yen and the US dollar in the lower panel, computed following Müller and Petalas (2010).

4.3.2. Deviating from the assumption of rational expectations: survey-based evidence

The focus of this paper is on the risk-characteristics of Swiss franc bilateral exchange rates, motivated by the idea that risk factors, together with interest rate differentials, can explain exchange rate movements ex-post. The regression framework is motivated by UIP under the assumption of rational expectations.

In this subsection we investigate whether UIP holds ex-ante for Swiss franc bilateral exchange rates, when the rational expectations assumption is dropped. To do this we use survey data from Consensus Economics. Every month, Consensus Economics asks a large number of financial analysts to provide exchange rate forecasts for horizons of 3 months ahead, 1 year ahead and 2 years ahead. We focus on 3-month ahead expectations. Surveys are conducted in the beginning of each month (typically, surveys are published around the 10th day of each month, with survey answers being received in the days prior). For the survey published in month t, expectations are for the end of month $t + 3$. We use only data for every fourth month to avoid using overlapping forecasts, which would likely induce additional serial correlation due to overlapping information sets. This leaves us with 38 observations for the sample from January 1999 to August 2011.

We run the following regression:

$$E_t\left(s_{t+1}^k\right) - s_t^k = \alpha^k + \beta^k\left(f_t^k - s_t^k\right) + \varepsilon_{t+1}^k \tag{14}$$

where the forward discount is measured for the day on which the Consensus survey is published.[10] The results are reported in Table 4.

UIP is seen to hold ex-ante for most currency pairs in the sense that (1) expectations are unbiased, since the constant is typically not statistically significantly different from zero; and (2) the coefficient on the forward discount is typically positive. With the exception of the Japanese yen and the Swedish krona, the results therefore provide strong support that UIP holds ex-ante, when measured with survey expectations. The coefficients on the forward discount are, however, clearly different from 1.

Note that even though non-overlapping forecasts are used, the residuals still exhibit serial correlation as shown by Durbin–Watson statistics well below 2. While we use Newey–West corrected standard errors to assess statistical significance, the low values of the Durbin Watson together with the relatively small number of observations suggest that the results should be interpreted with caution.

4.3.3. Other robustness checks

To check the robustness of our baseline specification we conduct a number of additional analyses summarized below. To conserve space, we do not report the detailed results but all of them are available upon request.

First, we add the spread (in levels or changes) between Italian and German 10-year government bond yields as a control variable for specific euro area risks to the baseline regression setup. Since the Swiss economy is closely related to the euro area, one might argue that it is not necessarily Swiss-specific or global risk but the materialization of euro area risks that influences Swiss franc exchange rate changes. The Italian–German bond yield spread appears to be a good proxy for these risks. The spread is particularly high in the early to mid-1990s when the European exchange rate mechanism experienced a crisis episode and in the recent past reflecting the euro area sovereign debt issues. It turns out that the regression coefficients of the bond spread in almost all regressions (even in the euro regression) are not statistically significant. Including the bond spread does not change any of the estimates of the baseline specification and does not improve the regression fit.

[10] Unfortunately it is impossible to measure the forward discount and expectations precisely on the same day, since financial analysts may submit their forecasts to Consensus Economics on different days. Most likely our forward discount is measured shortly after exchange rate expectations are formed.

Table 3
EURCHF in augmented UIP regressions.

	α^k (Constant)	β_0^k $(f_t^k - s_t^k)$	β_1^k (AFX_{t+1})	β_2^k (*100) $(\Delta \log(\text{VIX})_{t+1})$	R^2	DW
Panel A: January 1990 to August 2011 (ECU and euro data)						
ECU/EURO	− 0.00 (0.00)	− 0.95 (0.93)	0.35* (0.05)	−0.75 (0.49)	0.33	2.19
Panel B: January 1999 to August 2011 (euro only)						
EURO	− 0.00* (0.00)	− 2.09 (1.10)	0.43* (0.08)	−1.44* (0.65)	0.40	2.36
Panel C: January 1999 to December 2009						
EURO	0.00 (0.00)	0.48 (1.35)	0.30* (0.06)	− 1.32* (0.57)	0.38	2.22
Panel D: January 1999 to June 2007						
EURO	0.00 (0.00)	1.29 (1.29)	0.19* (0.04)	− 0.95* (0.43)	0.23	2.06

Notes: This table presents estimates from regressions of Swiss franc exchange rate changes against the ECU/euro on a constant, the respective forward discount and two contemporaneously measured risk factors, i.e.

$$\Delta s_{t+1}^k = \alpha^k + \beta_0^k \left(f_t^k - s_t^k\right) + \beta_1^k \text{AFX}_{t+1} + \beta_2^k \Delta \log(\text{VIX})_{t+1} + \varepsilon_{t+1}^k$$

with AFX abbreviating average Swiss franc exchange rate returns excluding the ECU/euro and Δ log(VIX) representing log changes in the CBOE option-implied volatility index, VIX. *s* and *f* represent log spot exchange rate and 1-month log forward rates respectively. Newey–West (Newey and West, 1987) corrected standard errors appear below the point estimates in parentheses. The R^2 statistic is adjusted for the number of regressors. An asterisk indicates that estimates are significantly different from zero at 95% confidence level. We report the Durbin–Watson statistic for the autocorrelation of the regression residuals under the column "DW".
Panel A gives the results when combining ECU and euro data to extend the sample period back to January 1990. The sample period ends in August 2011. Panel B shows the results when we use euro data only. Accordingly the sample period runs from January 1999 to August 2011. To gauge the impact of the recent crisis periods on the regression results, Panel C reports the estimates for the sample period from January 1999 to December 2009, i.e. the sample end before the intensification of the euro area sovereign debt crisis. Finally, panel D provides the corresponding estimates for a sample ending in June 2007 thus excluding both the global financial crisis and the sovereign debt crisis in the euro area.

Second, we assess if our baseline results hold once we include a measure of funding liquidity risk. We use the TED spread (in levels or changes), proposed by Brunnermeier et al. (2009), for that purpose. The TED spread is commonly thought of as a measure of investors' funding liquidity risk. If the spread is low there is little funding liquidity risk, while a large and negative spread indicates funding liquidity risk. Brunnermeier et al. (2009) show that the TED spread is a good explanatory variable for carry trade returns. Although we find significant comovement between Swiss franc exchange rate returns and the TED spread, none of the baseline specification estimates is altered.

Table 4
UIP regressions (based on survey data).

	α^k	β^k	R^2	DW
CAD	0.01 (0.01)	3.47* (1.28)	0.16	1.63
EUR	0.01* (0.01)	2.04* (1.00)	0.11	1.02
JPY	0.00 (0.01)	− 3.82* (0.82)	0.17	1.46
NOK	0.02* (0.01)	8.57* (4.02)	0.13	1.21
ZAR	− 0.01 (0.01)	4.70* (1.67)	0.07	2.00
SEK	0.01 (0.01)	− 6.53 (9.79)	0.02	1.22
GBP	0.02* (0.01)	1.62* (0.30)	0.42	1.34
USD	0.01 (0.01)	3.12* (0.61)	0.40	1.34

Notes: This table presents estimates from regressions of expected Swiss franc exchange rate returns (Swiss franc relative to Australian dollar, Canadian Dollar, euro, Japanese yen, New Zealand dollar, Norwegian krone, Singaporean dollar, South African rand, Swedish krona, British pound and US dollar) on a constant and the respective forward discount, i.e.

$$\Delta s_{t+1}^k = \alpha^k + \beta^k \left(f_t^k - s_t^k\right) + \varepsilon_{t+1}^k$$

where *k* denotes the currency pairs and *s* and *f* represent log spot exchange rates and 3-month log forward rates, respectively. Newey–West (Newey and West, 1987) corrected standard errors appear below the point estimates in parentheses. The R^2 statistic is adjusted for the number of regressors. An asterisk indicates that estimates are significantly different from zero at 95% confidence level. We report the Durbin–Watson statistic for the autocorrelation of the regression residuals under the column "DW". The sample period runs from January 1999 to August 2011. Exchange rate expectations are obtained from Consensus Economics, and a large number of financial analysts are included in the survey. Consensus Economics collects forecasts in the beginning of each month (survey data is typically published around the 10th day of the month); we use 3-month ahead forecasts, i.e. forecasts formed at the beginning of month t for the end of month $t + 3$. Consensus forecasts are available monthly, but we use only observations of every fourth month (38 observations) to avoid using overlapping forecasts, which would induce serially correlated errors.

As a third robustness check, we use the level of VIX instead of changes in VIX in the augmented UIP regressions. For instance, Ranaldo and Söderlind (2010) use the level of VIX in their high frequency assessment of safe haven currencies as one potential risk factor. Interestingly, we find only little covariation between monthly exchange rate returns and the level of VIX. At this relatively low frequency returns in the VIX index seem to be a better proxy of global risk, which is in line with Lustig et al. (2011) and the finding that their global risk factor is highly correlated with innovations in global equity market volatility rather than with volatility levels.

Finally, we follow Engel et al. (2012) and extract principal components from the Swiss franc exchange rates under study. We then use these principal components as proxy of risk factors. Since a direct link between exchange rate changes and macroeconomic aggregates is difficult to detect in the data, Engel et al. (2012) argue that the principal components could be interpreted as reflection of fundamental determinants of exchange rate changes. In our context, the first three principal components explain about 80% of the common variation in the Swiss franc exchange rate changes. On the first principal component all Swiss franc exchange rates load positively. It is closely correlated with the average exchange rate return. Interestingly, the third principal component could reflect carry trade risk as we observe clear differences in the loading on that principal component between typical funding currencies such as the Yen and US dollar and investment currencies such as the Australian dollar or the South African rand. Regressions of Swiss franc returns on the three principal components provide a slightly better fit, in terms of R^2 statistics, than our baseline specification. However, this better fit comes at the cost of an unclear economic interpretation of the principal components, which is particularly the case for the second principal component.

5. Conclusions

This paper estimates an asset pricing model relating bilateral Swiss franc exchange rate returns to alternative risk factors. We also explore how the influence of the risk factors changes over time, in particular between risk episodes and tranquil periods. We find that the Swiss franc does not behave like a safe haven asset against *all* currencies. The US dollar, the yen and the pound provided a better hedge against global risk than the Swiss franc during our sample period from January 1990 to August 2011. This finding could reflect the limited size of Swiss franc-denominated asset markets and hence limited market liquidity. Nevertheless, we find that the Swiss franc appreciates against many currencies when measures of global risk increase. This is the case both on average and in particular in times of financial stress.

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